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Part A:

1. How many duplicate transactions were removed?

85 duplicate transactions were removed.

2. How many invalid quantity records were removed?

120 invalid quantity records were removed.

3. How many records remain after cleaning?

3000 records remain after cleaning

4. What is the total revenue?

\$ 941, 694.59

5. What is the total gross profit?

\$309,308.93

Summary Findings

After cleaning the sales and store datasets, I analyzed the company's [performance to better understand which areas could be improved. ShopSmart Retail's cleaned sales data shows that revenue is \$ 941, 694.59 and gross profit is \$309,308.93 across 3,000 cleaned records. Although the business is doing well overall, there are some regions, and categories that make more revenue than others. There are areas for ShopSmart to improve its profits by focusing on weaker areas of the business.

Some Key Findings

1. The revenue by product category graph shows that electronics is making significantly more revenue than the other product categories. It accounts for 58.7% of the company's total revenue which is much higher than how much all the other product categories make. Although this is good that Electronics brings in a large sum of money for the company, this also does mean that ShopSmart should not heavily depend on one product category because if sales were to decrease in the future the company's overall revenue would significantly be affected. Therefore it is important to focus on growing other product categories as well to reduce the risk.

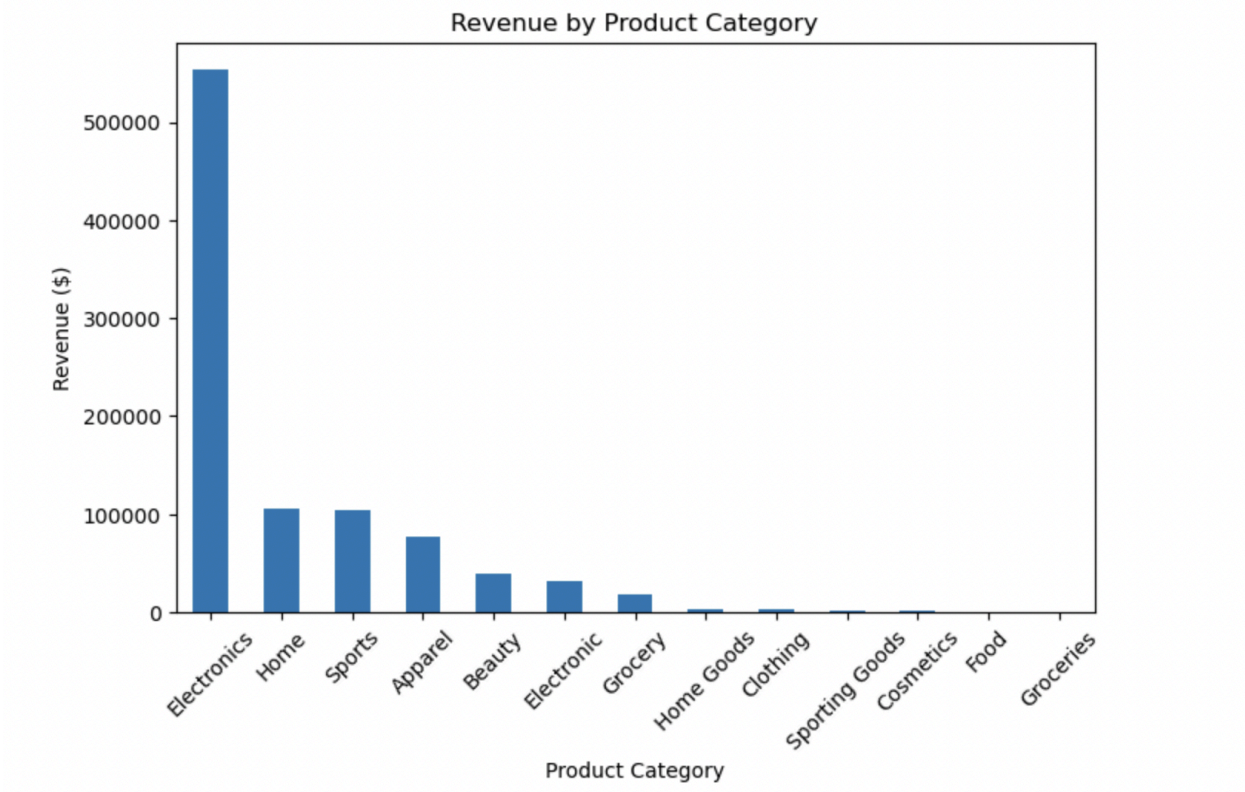
2. Additionally although electronics earns the most revenue, other categories such as Home, and Sports also contribute to the company's gross profit. Management should compare revenue and profit before deciding which categories to expand.
3. The analysis also showed that regional performance is uneven and it shows that the South region generated the highest gross profit followed by the West and East and then the North region with the lowest gross profit which is significantly lower than South as shown on the graph. This could mean that stores in this region have lower sales, higher operating costs, or lower customer demand. This region would need to get attention from the management team to better understand why it is not making as much gross profit.

Based on the analysis I would recommend:

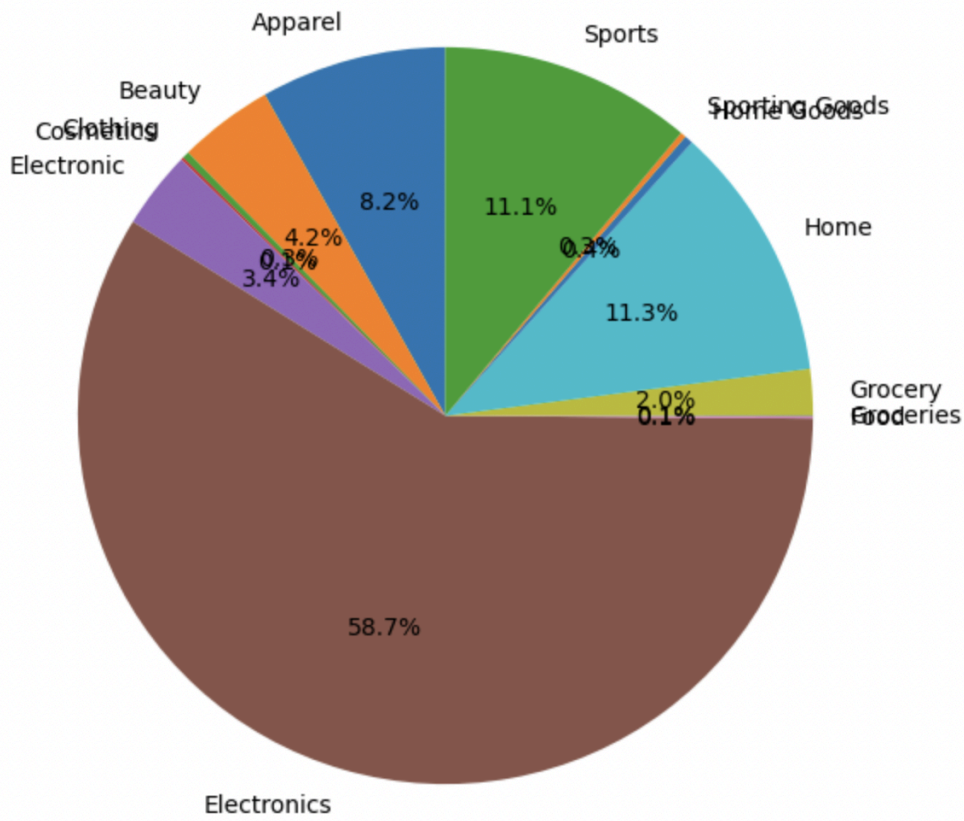
- 1) Continue investing in the product category electronic because it generates the largest share of revenue. Simultaneously try to grow other profitable categories so the company is not heavily reliant on electronics for its profits. I would suggest continuing focusing on Home, Sports, Apparel and Beauty since those areas are making profit currently therefore also focusing on those can help reduce the reliance on Electronics.
- 2) Look into why the North region has the lowest growth profit. The Management team should focus on sales, inventory, staffing and customer demand to better understand what is causing the low gross profit from that region. They can try and improve the West and East region as well.

Final Conclusion

To conclude after the data cleaning process I found that the company is successful overall with strong revenue and healthy gross profit. However the company does heavily rely on the Electronic category and experiences uneven performance across its regions. By improving these two areas the company can reduce business risk, and increase profit in low performing regions.



Revenue Share by Product Category



Gross Profit by Region

